Aditya Mittal

Backend Engineer | Distributed Systems
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EDUCATION

Jaypee University of Information Technology **2023 – 2027**
B.Tech – Computer Science and Engineering (Minor: ECE) *Solan, Himachal Pradesh*

EXPERIENCE

Basepair **June 2026 – August 2026**
Software Development Engineer in Test (SDET) Intern *Remote*

- Diagnosed and resolved critical production **distributed systems** defects, including an **OAuth race condition** via DB-level row locking and multi-instance file-access failures via **AWS S3** on-demand fallbacks.
- Led a full-stack **SonarQube remediation** initiative across Python/Django and React codebases, resolving 100+ BLOCKER/HIGH security and reliability defects (SQL injection, ReDoS, insecure crypto, unhandled exceptions) and authoring organizational **Standards.md** guidelines.

Spelll Production **June 2024 – July 2024**
Frontend Engineering Intern *Remote*

- Owned the end-to-end frontend lifecycle from prototyping to production deployment, translating UI/UX mockups into pixel-perfect, mobile-first components and integrating client-side form validation to improve engagement and lead conversion.

PROJECTS

MedVault | Django, DRF, PostgreSQL, Gemini API, React, JWT | Live Demo **Feb 2026 – Aug 2026**

- Engineered a structured **Google Gemini API prompting pipeline** with a hard-coded JSON output schema and explicit safety constraints preventing diagnosis/prescription hallucinations, capping input at 50,000 characters and adding a tolerant **JSON-from-prose fallback parser** for malformed model output.
- Designed a **Django REST Framework** backend with a custom email-based User model, **JWT authentication with refresh-token rotation**, and per-user data isolation enforced at the queryset level; isolated all Gemini calls behind a single mockable client so a **47-test backend suite** runs with zero API key configured.

Canopy | Django, PostgreSQL, tree-sitter, Gemini API, Plotly Dash | Live Demo **Aug 2026 – Sep 2026**

- Designed a **multi-language static-analysis engine** unifying Python's ast module and **tree-sitter** behind one generic analyzer interface, supporting 10 languages with deterministic **cross-file call-graph resolution**.
- Architected an optional AI chat feature fully isolated from the deterministic analysis pipeline — grounded only in already-computed call-graph results, keeping the core parsing product 100% LLM-free with zero hallucination risk — backed by a **commit-hash-keyed PostgreSQL caching layer** and least-privilege, **zero-scope GitHub OAuth**.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, C, C++, SQL
Backend & Frameworks: Node.js, Express.js, Django, Django REST Framework, JWT, OAuth 2.0
Cloud & Infrastructure: AWS CloudFormation, AWS (S3, EC2, Lambda, RDS), Docker, Kubernetes
Databases: PostgreSQL, MongoDB
DevOps & Tooling: Docker Compose, Git, GitHub, Postman, Linux/Bash, CI/CD

EXTRACURRICULAR

Technical Sciences and Movies Photography Club (TMP), Member (2023–2025) — organized tech events, coding contests, workshops; mentored peers.
Omega LEO Club, Member (2024–2025) — led community service initiatives.